

# Product specification: Offline construction photo tracker

## 1. Instructions for the development agent

Design and implement an Android application for structured photographic tracking of a single large construction project.

The application will be used by a civil engineer while physically walking through a construction site. The engineer must be able to navigate through the construction hierarchy, select a service, capture multiple photos and later generate a structured PDF report.

The main purpose of the application is to eliminate the current manual workflow of:

1. Taking construction photos with a phone.
2. Moving the photos to a computer.
3. Renaming every file manually.
4. Separating photos into folders.
5. Organizing photos by quadra, prédio, pavimento, unidade and serviço.
6. Manually creating PDF reports.

The application must work completely offline. All structured data and original photos must remain on the Android device.

Do not introduce cloud services, authentication, synchronization, web APIs or online dependencies into version 1.

All user-facing content must be written in Brazilian Portuguese.

---

## 2. Working product definition

### 2.1 Suggested working name

**Registro Fotográfico de Obra**

This is only a working name and must be easy to replace later.

### 2.2 Product summary

Registro Fotográfico de Obra is an offline Android application that allows a civil engineer to:

- Configure the physical structure of a large construction project.
- Navigate through Quadra, Prédio, Pavimento and Unidade.
- Select a Serviço.
- Capture or import multiple photos.

- Automatically associate every photo with its complete construction hierarchy.
- Add a visible date and time watermark to every photo.
- Review previous photographic inspections by date.
- Generate a PDF report for one Quadra and one date.
- Export a PDF alone or a ZIP containing the PDF and all corresponding photos organized in folders.
- Monitor local storage consumption.
- Remove old photographic records by date.

The application is a photo-first construction documentation tool. It is not a complete construction management, project management or RDO platform.

---

## **3. Target user and usage environment**

### **3.1 Primary user**

A civil engineer or construction inspector working directly at the construction site.

### **3.2 Typical environment**

The user may be:

- Walking between buildings and apartments.
- Using the application with one hand.
- Wearing gloves.
- Working under strong sunlight.
- Working in dusty areas.
- Moving through locations with no cellular or Wi-Fi connection.
- Taking many photos in a short period.
- Repeatedly moving between neighboring units.
- Using a phone with limited available storage.

### **3.3 UX consequences**

The interface must:

- Prioritize speed over decorative complexity.
  - Use large buttons and list items.
  - Avoid small icons as the only way to perform important actions.
  - Use strong contrast for outdoor visibility.
  - Minimize typing.
  - Preserve navigation state when returning to the previous level.
  - Allow repeated photo capture without reopening the hierarchy after each photo.
  - Clearly show the currently selected location.
  - Provide immediate visual feedback after a photo is saved.
  - Avoid gestures that are not visually discoverable.
  - Never depend on an internet connection.
-

## 4. Platform and technical constraints

### 4.1 Target platform

- Android only.
- Native Android application.
- Brazilian Portuguese locale.
- Portrait orientation should receive the highest design priority.
- Landscape orientation may be supported but is not the primary workflow.

### 4.2 Recommended Android stack

Use:

- Kotlin.
- Jetpack Compose.
- Material 3.
- Room for the local relational database.
- CameraX for in-app camera capture.
- AndroidX ExifInterface for image metadata.
- Android Photo Picker for importing photos from the gallery.
- Storage Access Framework for selecting export destinations.
- Kotlin coroutines for asynchronous operations.
- A background or foreground task mechanism for long report and ZIP generation operations.
- Dependency injection through a standard Android solution.
- MVVM or another clear unidirectional state architecture.

The application should not request the `INTERNET` permission in version 1.

Do not request broad access to all files on the device. Use the Android system picker for gallery import and export destinations.

### 4.3 Local source of truth

The Room database and local photo files are the only sources of truth.

The application must:

- Start and operate normally in airplane mode.
  - Never wait for a network request.
  - Never display synchronization states.
  - Never require a user account.
  - Never require a server.
-

## 5. Product scope

### 5.1 Included in version 1

- One construction project.
- Structure configuration.
- Quadras.
- Prédios.
- Pavimentos.
- Unidades.
- Unit types.
- Serviços.
- Bulk structure generation.
- Pavimento duplication.
- Prédio duplication.
- Check-ins or inspection sessions.
- Camera capture.
- Gallery import.
- Photo watermark.
- Local photo storage.
- Historical inspection visualization.
- PDF generation.
- ZIP generation with organized folders.
- PDF and ZIP sharing.
- Storage consumption visualization.
- Deletion of records by date.
- Renaming and archiving structure items.

### 5.2 Explicitly outside version 1

Do not implement:

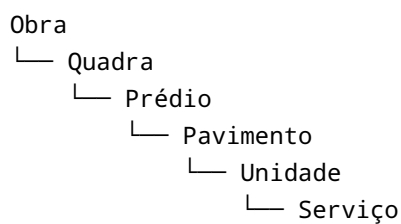
- Authentication.
- Multiple users.
- Multiple construction projects.
- Cloud backup.
- Cloud synchronization.
- Web dashboard.
- Photo notes.
- Photo annotations or drawing.
- Arrows or markings on images.
- Progress percentages.
- Service statuses.
- Before-and-after classification.
- Issues or nonconformities.
- Signatures.
- Weather.
- Materials.
- Labor tracking.
- Equipment tracking.

- Videos.
- GPS tracking.
- Notifications.
- Approval workflows.
- Automatic image recognition.
- Restore from backup.
- Report templates configurable by the user.

---

## 6. Terminology and construction hierarchy

Use these exact Brazilian Portuguese concepts throughout the interface:



### 6.1 Obra

The application manages one Obra.

Required field:

- Nome da obra.

Optional fields:

- Endereço.
- Empresa.
- Responsável técnico.

These optional fields may appear in the PDF header when present.

### 6.2 Quadra

A Quadra is the largest navigable construction division below the Obra.

Examples:

- Quadra A.
- Quadra B.
- Quadra 01.

## 6.3 Prédio

A Prédio belongs to one Quadra.

Examples:

- Prédio 01.
- Torre A.
- Edifício B.

Use the interface term **Prédio**, but allow the user to choose any name.

## 6.4 Pavimento

A Pavimento belongs to one Prédio.

Examples:

- Subsolo 02.
- Subsolo 01.
- Térreo.
- 1º Pavimento.
- 2º Pavimento.
- Cobertura.

Each Pavimento must contain:

- Display name.
- Sort order.

Do not depend exclusively on its name for sorting. The user must be able to order Pavimentos correctly even when using names such as Térreo, Mezanino and Cobertura.

## 6.5 Unidade

Use **Unidade** as the generic hierarchy term.

A Unidade can represent:

- Apartamento.
- Garagem.
- Academia.
- Salão de festas.
- Área comum.
- Área técnica.
- Loja.
- Depósito.
- Cobertura.
- Hall.
- Escada.

- Elevador.
- Outro.

Each Unidade must contain:

- Nome.
- Tipo.
- Pavimento.
- Sort order.
- Active or archived state.

Examples:

- Unidade 101, type Apartamento.
- Vaga G01, type Garagem.
- Academia, type Área comum.
- Hall social, type Hall.

The type list should include common predefined options and allow a custom type named **Outro**.

## 6.6 Serviço

A Serviço is a reusable label selected when photos are taken.

Examples:

- Hidráulica.
- Elétrica.
- Alvenaria.
- Pintura.
- Revestimento.
- Impermeabilização.
- Esquadrias.
- Forro.
- Instalação de louças.
- Acabamento.

Services should be configured once at the Obra level. Do not require the user to manually create the same services inside every Unidade.

During photo capture, a photo record associates:

- One Unidade.
- One Serviço.

This preserves the visible hierarchy:

Quadra > Prédio > Pavimento > Unidade > Serviço

without duplicating hundreds of service records during initial configuration.

---

## 7. Main application navigation

Use a bottom navigation bar with four main destinations:

1. **Registrar**
2. **Histórico**
3. **Relatórios**
4. **Estrutura**

A settings or overflow action may be placed in the top application bar for:

- Dados da obra.
- Armazenamento.
- Informações do aplicativo.

### 7.1 Default destination

Open the application on **Registrar**.

The engineer's primary job is taking photos, so the capture workflow must be immediately accessible.

---

## 8. Module: initial configuration

### 8.1 First-use experience

On the first application launch, display a short setup flow:

1. Enter the Obra name.
2. Create the first Quadra.
3. Create or generate the first Prédio structure.
4. Add Serviços.
5. Finish setup and start registering photos.

Do not force the user to configure the entire construction before using the application. The structure must remain editable later.

### 8.2 Initial data warning

Because version 1 has no backup or restoration system, display a clear notice during onboarding:

Os dados e as fotos ficam armazenados somente neste dispositivo. A desinstalação do aplicativo ou a perda do aparelho poderá apagar os registros que não tenham sido exportados.

The warning must be understandable but should not block normal application use after confirmation.

---



## 9. Module: structure management

The Estrutura module must allow the engineer to view and configure:

- Obra.
- Quadras.
- Prédios.
- Pavimentos.
- Unidades.
- Serviços.

Use the same hierarchical navigation pattern used by the capture flow.

### 9.1 Basic operations

For Quadras, Prédios, Pavimentos, Unidades and Serviços, support:

- Create.
- Rename.
- Reorder.
- Archive.
- Restore archived item where appropriate.

### 9.2 Deletion rules

If an item has associated photographic records:

- Do not permit permanent deletion.
- Allow archiving.
- Show an explanation that the item cannot be deleted because it is used by existing records.

If an item has no records or children, permanent deletion may be allowed after confirmation.

### 9.3 Rename behavior

Renaming a Quadra, Prédio, Pavimento, Unidade or Serviço must update how existing records are displayed.

Do not rename physical image files internally whenever a structure label changes.

The exported paths and filenames must always use the latest current hierarchy names.

### 9.4 Archive behavior

Archived items:

- Must remain connected to historical records.
- Must appear in old check-ins and reports.
- Must not appear in the normal capture selection flow.

- May be displayed through an “Itens arquivados” option in Estrutura.
- 

## 10. Module: bulk structure creation

Bulk structure creation is a core requirement, not an optional enhancement.

A building may contain 23 to 30 Pavimentos and four to six apartments on each Pavimento. The user must not be required to create every Pavimento and Unidade manually.

### 10.1 Building creation options

When creating a Prédio, offer:

- Criar vazio.
- Gerar estrutura.
- Duplicar prédio existente.

### 10.2 Structure generator

Create a wizard named **Gerar estrutura do prédio**.

#### Step 1: Pavimentos

Allow the user to configure:

- First Pavimento number.
- Last Pavimento number.
- Pavimento name pattern.
- Sort direction.
- Optional special Pavimentos.

Example:

```
Do pavimento: 1
Até o pavimento: 23
Nome: {n}º Pavimento
```

Preview:

```
1º Pavimento
2º Pavimento
3º Pavimento
...
23º Pavimento
```

Special Pavimentos such as Subsolo, Térreo, Mezanino and Cobertura should be added separately.

## Step 2: Units per Pavimento

Allow configuration of:

- Number of Unidades per Pavimento.
- Unit type.
- First unit position.
- Numbering pattern.
- Optional prefix.
- Optional suffix.
- Number padding.

Supported placeholders should include:

- {pavimento}
- {unidade}
- {unidade:02}
- {sequencia}

Example configuration:

```
Pavimentos: 1 a 23
Unidades por pavimento: 4
Tipo: Apartamento
Formato: {pavimento}{unidade:02}
```

Preview:

```
Pavimento 1:
Apartamento 101
Apartamento 102
Apartamento 103
Apartamento 104

Pavimento 2:
Apartamento 201
Apartamento 202
Apartamento 203
Apartamento 204
```

The generator must support common Brazilian apartment numbering patterns.

## Step 3: Preview and confirmation

Before creating records, display:

- Number of Pavimentos to create.
- Number of Unidades to create.
- Sample of the first Pavimento.

- Sample of the last Pavimento.
- Warning for duplicated names.
- Final confirmation button.

All generated records must be created in a single database transaction. If generation fails, do not leave a partially created structure.

## 10.3 Pavimento model

Support a concept called **Pavimento tipo**.

Workflow:

1. User creates one Pavimento with its Unidades.
2. User chooses "Replicar pavimento".
3. User selects a target Pavimento range.
4. The app copies the unit layout to every selected Pavimento.
5. Unit names may be automatically renumbered according to the configured pattern.

This supports buildings where most floors have the same apartment arrangement.

## 10.4 Duplicate Pavimento

From a Pavimento options menu, provide:

- Duplicar para outro pavimento.
- Replicar para intervalo.

Only copy:

- Unit names or numbering rules.
- Unit types.
- Unit ordering.

Never copy:

- Photos.
- Check-ins.
- Historical records.

## 10.5 Duplicate Prédio

From a Prédio options menu, provide **Duplicar prédio**.

The workflow must:

1. Ask for the destination Quadra.
2. Ask for the new Prédio name.
3. Show the number of Pavimentos and Unidades that will be created.
4. Copy the complete Pavimento and Unidade structure.
5. Preserve ordering.

6. Exclude all photos, check-ins and historical records.

Example:

```
Origem: Quadra A > Prédio 01
Novo prédio: Quadra A > Prédio 02

Serão copiados:
23 pavimentos
92 unidades
```

Services do not need to be copied because they are reusable labels belonging to the Obra.

---

## 11. Module: check-ins

### 11.1 Definition

A **check-in** is a dated field inspection session.

One check-in may contain multiple photographic records from:

- Multiple Quadras.
- Multiple Prédios.
- Multiple Pavimentos.
- Multiple Unidades.
- Multiple Serviços.

The user should not need to create a new check-in for every apartment.

### 11.2 Starting a check-in

In Registrar, show:

- **Iniciar vistoria**
- **Continuar vistoria de hoje**, when an unfinished or recent session exists.

When starting a session, store:

- Start date and time.
- End date and time, when finalized.
- Device time-zone offset.

No title or description is required.

### 11.3 Check-in completion

The engineer may finish a check-in manually.

Finishing a check-in must not prevent adding another check-in on the same day.

A day may contain multiple check-ins.

Reports are based on photo capture date, not exclusively on the check-in start date.

---

## 12. Module: photo registration

### 12.1 Required navigation flow

The primary capture flow is:

```
Registrar
→ Quadra
→ Prédio
→ Pavimento
→ Unidade
→ Serviço
→ Câmera/Fotos
```

This is the required Option A flow.

### 12.2 Hierarchy selection screens

Each level should use a clear full-width list.

Each item may show:

- Name.
- Type, where applicable.
- Number of children.
- Number of photos taken today, when available.

Examples:

```
Quadra A
4 prédios · 126 fotos hoje
```

```
2° Pavimento
6 unidades
```

```
Apartamento 204
12 fotos hoje
```

## 12.3 Breadcrumb

After the user selects a Quadra, show a persistent context breadcrumb.

Example:

Quadra A > Prédio 01 > 2º Pavimento > Apartamento 204

On smaller screens, allow the breadcrumb to wrap or horizontally scroll.

Never hide the selected location while the user is choosing a Serviço or taking photos.

## 12.4 Back navigation

The Android back action must return exactly one level while preserving:

- The previous scroll position.
- The previous selected item.
- The active check-in.
- The current date.
- The last visited hierarchy.

Example:

Câmera → Serviços → Unidade → Pavimento

## 12.5 Recent and continuation shortcuts

To reduce repeated navigation, Registrar may show:

- Continuar de onde parei.
- Últimas unidades acessadas.
- Último serviço utilizado.

These are shortcuts only. They must not replace the full hierarchy.

## 12.6 Service selection

The service screen displays the active Serviço labels.

The user selects one service and enters the photo capture screen.

Do not ask for:

- Status.
- Progress.
- Notes.

- Descriptions.
- Completion percentage.

## 12.7 Capture screen

The capture screen must show:

- Camera preview.
- Current hierarchy.
- Current service.
- Large shutter button.
- Gallery import button.
- Thumbnail strip of photos already captured in this photographic group.
- Number of photos captured.
- Back button.
- Button to change service.
- Optional shortcut to return to the Unidade list.

The shutter button should be large and reachable with one hand.

After a successful capture:

- Keep the camera open.
- Display the new thumbnail immediately.
- Provide haptic or subtle visual confirmation.
- Allow another photo to be taken immediately.
- Do not return to the hierarchy automatically.

## 12.8 Correction during capture

The engineer must be able to delete a newly captured photo from the thumbnail preview.

Ask for confirmation before deletion.

This is necessary to remove accidental, blurred or incorrect photos.

---

# 13. Photo import from gallery

Allow the user to select one or multiple photos through Android's system photo picker.

Imported photos must be copied into the application's local controlled storage.

For each imported photo:

1. Read the original EXIF capture date and time when available.
2. Use that timestamp as the photo capture date.
3. If the original timestamp is unavailable, use the import date and time.
4. Display a brief message when the import time had to be used.



5. Apply the visible watermark.
6. Write or preserve EXIF date and time information in the processed image.
7. Associate the image with the currently selected hierarchy and Serviço.

Do not keep a permanent dependency on the original gallery URI after the import has completed.

---

## 14. Photo watermark and metadata

### 14.1 Visible watermark

Every stored construction photo must contain a visible date and time watermark.

Recommended content:

27/06/2026 14:35  
Quadra A · Prédio 01 · 2º Pavimento  
Apartamento 204 · Hidráulica

The first line must always contain:

- Date in `dd/MM/yyyy`.
- Time in `HH:mm`.

The remaining lines identify:

- Quadra.
- Prédio.
- Pavimento.
- Unidade.
- Serviço.

### 14.2 Watermark placement

Default position:

- Bottom-left corner.

Requirements:

- Semi-transparent dark background.
- White text.
- Sufficient internal padding.
- Safe margin from image edges.
- Maximum of three compact lines.
- Text size calculated according to image dimensions.
- Do not cover an excessive portion of the photo.
- Preserve readability on light and dark backgrounds.
- Respect image orientation.

The watermark is mandatory and cannot be disabled in version 1.

## 14.3 Canonical capture time

Use one canonical `capturedAt` value for:

- Database record.
- Watermark.
- EXIF original date.
- Report filtering.
- Export filename.

This prevents disagreements between the visible timestamp and stored data.

Store:

- Timestamp.
- Local timezone offset.
- Import timestamp separately when the source is the gallery.

## 14.4 Internal image processing

When processing a camera photo:

1. Capture to a temporary file.
2. Read and correct orientation.
3. Apply the watermark.
4. Write required EXIF timestamp fields.
5. Save the completed image to permanent local storage.
6. Generate a smaller thumbnail.
7. Insert the database record.
8. Delete the temporary file.

If any critical stage fails, do not insert an incomplete photo record.

---

# 15. Internal photo storage

## 15.1 Storage strategy

Store application-managed images in app-specific persistent storage.

Use:

- Original-quality watermarked image.
- Small thumbnail for lists and history.
- Temporary optimized copies when generating PDFs.

Do not load full-resolution images into list screens.

## 15.2 Internal filenames

Use stable unique internal filenames, such as UUIDs.

Example:

```
9f5cf8c4-46ac-4a8c-bbda-f6038774c194.jpg
```

Do not use the complete construction path as the internal filename.

This prevents file breakage when the user renames a Quadra, Prédio, Pavimento, Unidade or Serviço.

Human-readable paths and filenames are generated only during export.

## 15.3 Storage validation

Before capturing or importing photos:

- Check available storage.
- Display a warning when storage is low.
- Prevent capture only when saving the photo would be unsafe.

---

# 16. Module: history

## 16.1 Main historical view

The Histórico screen must group records by date, with the newest date first.

Each date card should show:

- Date.
- Number of check-ins.
- Number of photos.
- Number of Quadras represented.
- Approximate storage used.

Example:

```
27/06/2026  
2 vistorias · 184 fotos · 3 quadras · 1,8 GB
```

## 16.2 Date detail

Opening a date should display:

Data

- Quadras
- Prédios
- Pavimentos
- Unidades
- Serviços
- Fotos

Only show branches containing photos from that date.

## 16.3 Photo viewer

The photo viewer must support:

- Full-screen photo.
- Visible hierarchy.
- Serviço.
- Capture date and time.
- Swipe between photos.
- Delete individual photo with confirmation.

No notes or editing tools are required.

## 16.4 Empty states

Use clear empty-state messages.

Example:

Nenhum registro fotográfico foi realizado nesta data.

Do not use technical database terminology.

---

# 17. Module: PDF reports

## 17.1 Report filters in version 1

The report generator must require exactly:

- One date.
- One Quadra.

No other report filters are required in version 1.

The generated report includes all photos matching:

```
captured date = selected date
AND
Quadra = selected Quadra
```

All matching child records must be included automatically.

## 17.2 Report generation flow

The Relatórios screen should provide:

1. Select date.
2. Select Quadra.
3. Display summary.
4. Generate report.
5. Select export option.

The date picker should visually indicate dates that contain photos when practical.

After selecting the date, the Quadra list should show the number of matching photos.

Example:

```
Quadra A – 126 fotos
Quadra B – 84 fotos
```

Disable Quadras that have no photos for the selected date, or omit them from the list.

## 17.3 Report summary

Before generation, show:

```
Data: 27/06/2026
Quadra: Quadra A
Prédios: 4
Unidades registradas: 38
Fotos: 126
```

## 17.4 PDF format

Use:

- A4 pages.
- Portrait orientation.
- Consistent margins.
- Two photo columns.
- Four to six photos per page.

- Page numbers.
- Project and report context in the header.

Recommended layout:

- Two columns.
- Up to three image rows.
- Six images when image proportions permit.
- Four images when additional vertical space is required.
- Maintain image aspect ratio.
- Do not stretch images.

## 17.5 PDF hierarchy

The PDF must organize content in this sequence:

```

Quadra
→ Prédio
→ Pavimento
→ Unidade
→ Serviço
→ Fotos

```

Example:

```

QUADRA A

PRÉDIO 01

2° PAVIMENTO

APARTAMENTO 201

HIDRÁULICA
[foto] [foto]
[foto] [foto]

ELÉTRICA
[foto] [foto]

APARTAMENTO 202

HIDRÁULICA
[foto] [foto]

```

Only include sections that contain photos.

Do not print empty Prédios, Pavimentos, Unidades or Serviços.

## 17.6 Sorting

Use deterministic natural sorting:

1. Prédio sort order.
2. Pavimento sort order.
3. Unidade sort order.
4. Serviço sort order.
5. Photo capture time.

Do not sort numbers exclusively as strings. For example, Unidade 10 must appear after Unidade 9, not after Unidade 1.

## 17.7 PDF header

The report header should contain:

- Nome da obra.
- Selected Quadra.
- Report date.
- Optional address.
- Optional company.
- Optional responsible engineer.

Keep the header compact. The primary report content is photographic.

## 17.8 Photo labels

The full hierarchy does not need to be repeated underneath every photo because:

- The section headings identify the hierarchy.
- The image watermark already contains the date, location and service.

A small capture time may be displayed beneath the image when layout space permits.

## 17.9 Pagination

When a service contains more photos than fit on one page:

- Continue the service on the next page.
- Repeat the current Unidade and Serviço heading in a compact format.
- Do not leave a heading alone at the bottom of a page.

## 17.10 PDF image optimization

Do not embed full-resolution camera files directly without processing.

Generate temporary PDF-specific image versions:

- Maintain aspect ratio.

- Use sufficient quality for inspection.
- Avoid excessive PDF file size.
- Avoid loading many full-size bitmaps into memory simultaneously.
- Process images incrementally.

Original full-quality images must remain unchanged.

## 17.11 PDF filename

Use a filename pattern such as:

```
Relatorio_2026-06-27_Quadra-A.pdf
```

Sanitize unsupported filename characters.

---

# 18. Module: exports

## 18.1 Export options

After generating a report, show:

1. **Salvar somente PDF**
2. **Salvar PDF e fotos em ZIP**
3. **Compartilhar PDF**
4. **Compartilhar ZIP**, after ZIP creation

## 18.2 Destination selection

Use the Android system document and directory picker.

The user must choose where the PDF or ZIP is saved.

Do not silently save files into an inaccessible internal directory and call that an export.

## 18.3 ZIP contents

When the user selects PDF and photos, generate one ZIP file.

Example:

```
Relatorio_2026-06-27_Quadra-A.zip
```

Contents:



```
Relatorio_2026-06-27_Quadra-A/  
├─ Relatorio_2026-06-27_Quadra-A.pdf  
└─ Fotos/  
    └─ Quadra A/  
        ├── Predio 01/  
        │   ├── 1 Pavimento/  
        │   │   ├── Apartamento 101/  
        │   │   │   ├── Hidraulica/  
        │   │   │   │   ├── 2026-06-27_09-15-32_001.jpg  
        │   │   │   │   └─ 2026-06-27_09-16-05_002.jpg  
        │   │   │   └─ Eletrica/  
        │   │   └─ Apartamento 102/  
        │   └─ 2 Pavimento/  
        └─ Predio 02/
```

## 18.4 Export image naming

Use:

```
YYYY-MM-DD_HH-mm-ss_NNN.jpg
```

Example:

```
2026-06-27_14-35-22_001.jpg
```

When two photos have the same capture timestamp, use the sequential suffix to avoid collisions.

## 18.5 Filename sanitization

For folder and file names:

- Remove path separators.
- Remove control characters.
- Replace unsupported symbols safely.
- Trim unnecessary spaces.
- Prevent duplicate path collisions.
- Preserve human readability.

Accented Portuguese characters may remain when supported by the ZIP implementation, but folder paths must never break due to special characters.

## 18.6 Export progress

PDF and ZIP generation may involve hundreds of photos.

Display:

- Progress indicator.
- Current stage.
- Photo count processed.
- Cancel action where safe.

Example:

```
Gerando arquivo...  
84 de 126 fotos processadas
```

Do not freeze the interface.

If generation is interrupted:

- Delete incomplete temporary files.
- Preserve all original records and photos.
- Allow the user to try again.

---

## 19. Module: storage management

### 19.1 Storage overview

Provide an Armazenamento screen containing:

- Total space used by the application.
- Space used by photos.
- Space used by thumbnails.
- Space used by temporary files.
- Total photo count.
- Oldest record date.
- Newest record date.

Example:

```
Armazenamento utilizado: 18,4 GB  
Fotos: 17,9 GB  
Miniaturas e dados: 500 MB  
Total de fotos: 2.842
```

### 19.2 Consumption by date

List dates with:

- Photo count.

- Storage used.
- Number of check-ins.
- Delete action.

Example:

27/06/2026  
184 fotos · 1,8 GB

### 19.3 Delete by date

The bulk cleanup operation must delete records by one selected date.

Before deletion, display:

Excluir registros de 27/06/2026?

184 fotos e 2 vistorias serão excluídas permanentemente.

Essa ação não pode ser desfeita.

Require explicit confirmation.

Deleting a date must remove:

- Photos captured on that date.
- Thumbnails.
- Photo records.
- Empty photographic groups.
- Check-ins that become empty and belong entirely to that date.
- Temporary generated files owned by the application for that date.

Do not remove:

- Quadras.
- Prédios.
- Pavimentos.
- Unidades.
- Serviços.
- Files previously exported to a user-selected external location.

### 19.4 Individual deletion

Individual photo deletion is allowed from:

- The current capture group.
- Historical photo viewer.

Bulk storage cleanup remains date-based.

---

## 20. Visual design system

### 20.1 Design direction

Use a practical industrial visual language:

- Clear.
- Durable.
- Professional.
- High contrast.
- Minimal decorative content.
- Easy to understand under pressure.
- Consistent with modern Android behavior.

Do not make the interface resemble a consumer social media photo gallery.

### 20.2 Suggested color palette

#### Primary

##### Azul profundo

#0D47A1

Use for:

- Primary buttons.
- Selected navigation.
- Active hierarchy.
- Important screen titles.
- Main progress indicators.

Use white text on this color.

#### Accent

##### Âmbar de obra

#F59E0B

Use sparingly for:

- Active inspection indicator.
- Important highlights.
- Selected dates.
- Storage warnings.

Do not use white text on bright amber. Use dark text.

## Background

#F5F7FA

## Surface

FFFFFF

## Primary text

#17202A

## Secondary text

#52606D

## Dividers and borders

#D9E2EC

## Success

#2E7D32

Use for:

- Photo saved.
- Report generated.
- Export completed.

## Error or destructive action

#B3261E

Use for:

- Deletion confirmation.
- Capture failure.
- Export failure.
- Insufficient storage.

## 20.3 Theme

Implement a polished light theme first.

Strong sunlight and construction-site usage make light surfaces and high-contrast text the preferred version 1 experience.

Dark mode is not required in version 1.

## 20.4 Typography

Use the Android system typeface or Roboto.

Recommended hierarchy:

- Screen title: 24sp, semibold.
- Section title: 20sp, semibold.
- List item title: 17–18sp, medium.
- Body: 16sp.
- Supporting text: 14sp.
- Buttons: at least 15–16sp, medium or semibold.

Do not use very light font weights.

## 20.5 Touch targets

All interactive elements must have a minimum touch area of 48dp by 48dp.

Prefer:

- 56dp or taller list rows.
- Large full-width hierarchy cards.
- Large camera controls.
- Adequate spacing between destructive and non-destructive actions.

## 20.6 Cards and lists

Use filled or outlined cards for primary hierarchy items.

A hierarchy item should have one large touch target rather than multiple small clickable zones.

Place secondary actions such as rename, archive and duplicate in a visible overflow menu.

## 20.7 Icons and labels

Important navigation actions must use both:

- Icon.
- Portuguese text label.

Avoid relying on icons alone for:

- Generate report.
- Export.
- Delete.
- Duplicate.
- Finish inspection.
- Change service.

## 20.8 Feedback

Use:

- Snackbar for successful operations.
- Dialogs for destructive confirmation.
- Inline error messages for invalid forms.
- Progress indicators for PDF and ZIP generation.
- Haptic feedback after photo capture when supported.

Avoid displaying raw exceptions or technical error messages.

---

# 21. Important screen definitions

## 21.1 Registrar home

Content:

- Current date.
- Start or continue inspection button.
- Continue from last location.
- Recent locations.
- Number of photos captured today.
- Active check-in indicator.

Primary action:

Iniciar vistoria

or:

Continuar vistoria

## 21.2 Quadra list

Content:

- Quadra cards.
- Photos captured today per Quadra.

- Search when the list becomes large.

## 21.3 Prédio list

Content:

- Current Quadra breadcrumb.
- Prédio cards.
- Today's photo count.

## 21.4 Pavimento list

Content:

- Current Quadra and Prédio.
- Ordered Pavimento list.
- Search or fast scrolling for buildings with many floors.

## 21.5 Unidade list

Content:

- Current Pavimento.
- Unit name.
- Unit type.
- Photo count today.
- Search.

## 21.6 Serviço list

Content:

- Current full hierarchy.
- Active services.
- Recently selected service may appear first, but preserve configured sort order elsewhere.

## 21.7 Camera

Content:

- Current hierarchy.
- Current Serviço.
- Camera preview.
- Shutter.
- Gallery import.
- Recent thumbnails.
- Photo count.
- Change service.
- Return to unit.



## 21.8 Histórico

Content:

- Dates grouped newest first.
- Photo count.
- Check-in count.
- Storage consumption.
- Date detail navigation.

## 21.9 Relatórios

Content:

- Date selector.
- Quadra selector.
- Report summary.
- Generate button.
- Recent locally generated reports may be shown as temporary convenience, but they are not the source of truth.

## 21.10 Estrutura

Content:

- Dados da obra.
  - Quadras.
  - Serviços.
  - Building structure management.
  - Bulk generator.
  - Archived items.
- 

# 22. Data model

Use a relational model similar to the following.

## 22.1 Project

```
Project
- id
- name
- address nullable
- company nullable
- responsibleEngineer nullable
- createdAt
- updatedAt
```

Only one Project record is active.

## 22.2 Block

```
Block
- id
- projectId
- name
- sortOrder
- archivedAt nullable
- createdAt
- updatedAt
```

## 22.3 Building

```
Building
- id
- blockId
- name
- sortOrder
- archivedAt nullable
- createdAt
- updatedAt
```

## 22.4 Floor

```
Floor
- id
- buildingId
- name
- numericReference nullable
- sortOrder
- archivedAt nullable
- createdAt
- updatedAt
```

## 22.5 UnitType

```
UnitType
- id
- name
- isSystemType
- sortOrder
```

## 22.6 Unit

```
Unit
- id
- floorId
- unitTypeId
- name
- numericReference nullable
- sortOrder
- archivedAt nullable
- createdAt
- updatedAt
```

## 22.7 Service

```
Service
- id
- projectId
- name
- sortOrder
- archivedAt nullable
- createdAt
- updatedAt
```

## 22.8 InspectionSession

```
InspectionSession
- id
- projectId
- startedAt
- finishedAt nullable
- timezoneOffset
- createdAt
```

## 22.9 PhotoGroup

A PhotoGroup represents one selected hierarchy and service within a check-in.

```
PhotoGroup
- id
- inspectionSessionId
- unitId
- serviceId
- createdAt
```

Block, Building and Floor can be resolved through Unit relationships.

## 22.10 Photo

```
Photo
- id
- photoGroupId
- internalFilename
- thumbnailFilename
- sourceType: CAMERA | GALLERY
- capturedAt
- importedAt nullable
- timezoneOffset
- width
- height
- orientation
- sizeBytes
- mimeType
- exifTimestampWritten
- watermarkVersion
- createdAt
```

## 22.11 GeneratedReport

Optional local tracking entity:

```
GeneratedReport
- id
- reportDate
- blockId
- photoCount
- generatedAt
- temporaryInternalFilename nullable
- status
```

Exports saved through the system picker should not be treated as managed internal files after completion.

---

## 23. Data integrity requirements

Implement:

- Foreign keys.
- Transactions for bulk generation.
- Transactions for building duplication.
- Stable IDs.

- Unique names within the same parent where appropriate.
- Normalized comparison to prevent accidental duplicate names differing only by spaces or letter case.
- Safe soft deletion through archiving.
- Referential integrity for historical photos.

A hierarchy item used in a historical record must never disappear from that historical view due to archiving.

---

## 24. Performance requirements

The application must remain usable with:

- Dozens of Quadras and Prédios.
- Hundreds of Pavimentos.
- Thousands of Unidades.
- Tens of thousands of photos.

Use:

- Lazy lists.
- Pagination or incremental loading where needed.
- Thumbnails.
- Database indexes.
- Background image processing.
- Incremental PDF rendering.
- Incremental ZIP writing.
- Natural sorting based on stored sort order.
- Efficient aggregate queries for counts by date.

Recommended database indexes include:

- Photo captured date.
- PhotoGroup session ID.
- PhotoGroup unit ID.
- PhotoGroup service ID.
- Building block ID.
- Floor building ID.
- Unit floor ID.
- Session start date.

Do not load every photo bitmap into memory during report generation.

---

## 25. Error handling

Handle these situations clearly:

## Camera permission denied

Message:

A permissão da câmera é necessária para tirar fotos. Você ainda pode selecionar imagens da galeria.

Provide a button to open application settings when permission was permanently denied.

## No available storage

Message:

Não há espaço suficiente para salvar novas fotos. Libere espaço no dispositivo ou exclua registros antigos.

## No photos for report

Message:

Não existem fotos para a data e a quadra selecionadas.

## Export canceled

Return to the report screen without losing the generated report configuration.

## Gallery image without timestamp

Message:

A foto não possui uma data original válida. Foi utilizada a data da importação.

## PDF or ZIP failure

Message:

Não foi possível gerar o arquivo. Nenhuma foto original foi alterada.

Provide a retry action.

## Duplicate structure name

Message:

Já existe um item com este nome neste local.

## Interrupted processing

Temporary files must be deleted on the next application start if they are no longer associated with an active operation.

---

## 26. Privacy and security

- All construction data remains local.
  - No analytics SDK is required.
  - No advertising SDK.
  - No remote crash reporting in version 1 unless explicitly approved later.
  - No background uploads.
  - No network permission.
  - Share files only after an explicit user action.
  - Expose files to other applications through secure Android content URIs.
  - Do not expose raw internal file paths.
- 

## 27. Accessibility and field usability

Implement:

- Large touch targets.
  - High text contrast.
  - Portuguese content descriptions for important icons.
  - Screen-reader-compatible labels.
  - Support for Android font scaling.
  - Do not communicate errors using color alone.
  - Keep hierarchy names visible as text.
  - Confirm destructive actions.
  - Ensure the camera capture button remains usable on common phone sizes.
  - Avoid very thin dividers and low-opacity text.
  - Avoid placing important controls too close to screen edges.
- 

## 28. Suggested development phases

### Phase 1: Foundation

Implement:

- Android project structure.
- Room database.
- Entities and relationships.
- Repositories.
- Offline-only configuration.

- Navigation.
- Material theme.

## **Phase 2: Structure**

Implement:

- Obra.
- Quadras.
- Prédios.
- Pavimentos.
- Unidades.
- Serviços.
- Rename, reorder and archive.
- Bulk structure generator.
- Pavimento duplication.
- Prédio duplication.

## **Phase 3: Photo capture**

Implement:

- Check-ins.
- Hierarchy navigation.
- CameraX.
- Gallery picker.
- Watermark.
- EXIF timestamps.
- Internal photo storage.
- Thumbnails.
- Individual photo deletion.

## **Phase 4: History**

Implement:

- Date grouping.
- Hierarchical historical navigation.
- Photo viewer.
- Counts and storage aggregation.

## **Phase 5: Reports and export**

Implement:

- Date and Quadra selection.
- PDF layout.
- Two-column photo grid.
- PDF pagination.
- ZIP folder structure.



- Android save picker.
- Sharing.
- Progress and cancellation.

## Phase 6: Storage and hardening

Implement:

- Storage overview.
  - Storage grouped by date.
  - Delete by date.
  - Low-storage behavior.
  - Temporary file cleanup.
  - Performance testing.
  - Large dataset testing.
- 

## 29. Acceptance criteria

The version 1 application is acceptable only when all the following scenarios work.

### 29.1 Offline operation

Given that the device is in airplane mode:

- The application opens.
- The structure can be created and edited.
- Photos can be taken.
- Gallery photos can be imported.
- History can be viewed.
- PDFs can be generated.
- ZIP files can be generated.
- Files can be saved locally.

### 29.2 Bulk generation

Given:

- One Prédio.
- Pavimentos 1 through 23.
- Four apartments per Pavimento.
- Pattern `{pavimento}{unidade:02}`.

The application creates:

- 23 Pavimentos.
- 92 Unidades.
- Correct names from 101–104 through 2301–2304.
- Correct ordering.

- No partial records if creation fails.

## 29.3 Building duplication

Given an existing Prédio with:

- 23 Pavimentos.
- 92 Unidades.

When the user duplicates it:

- The new Prédio has an independent copy of the complete structure.
- No photos are copied.
- No check-ins are copied.
- No historical records are copied.

## 29.4 Photo capture

Given the selected path:

Quadra A  
Prédio 01  
2º Pavimento  
Apartamento 204  
Hidráulica

When a photo is taken:

- It is stored locally.
- It is associated with the active check-in.
- It is associated with the selected Unidade and Serviço.
- It contains the visible date and time watermark.
- It contains the hierarchy in the watermark.
- Its EXIF date is preserved or written.
- It immediately appears in the thumbnail list.
- The camera remains open for the next photo.

## 29.5 Navigation

After taking photos:

- Back returns to Serviços.
- Back again returns to Unidade.
- Previous selections and list positions are preserved.
- The active check-in remains active.

## 29.6 History

For a selected date:

- Only locations containing photos from that date are displayed.
- The user can navigate down to individual photos.
- The hierarchy and capture time are visible.
- A mistaken photo can be deleted.

## 29.7 PDF report

Given:

- Date 27/06/2026.
- Quadra A.

The generated PDF:

- Contains only photos captured on 27/06/2026 in Quadra A.
- Groups content by Prédio, Pavimento, Unidade and Serviço.
- Uses two photo columns.
- Displays four to six photos per page.
- Maintains image aspect ratios.
- Omits empty hierarchy sections.
- Uses deterministic ordering.
- Contains readable watermarks.
- Includes page numbers.

## 29.8 ZIP export

The generated ZIP:

- Contains the PDF.
- Contains all corresponding original-quality watermarked photos.
- Uses the expected folder hierarchy.
- Uses unique timestamp-based filenames.
- Can be saved through the Android system picker.
- Can be shared after creation.

## 29.9 Storage deletion

When the user deletes one date:

- All photos from that date are removed.
- All thumbnails from that date are removed.
- Related empty session records are cleaned up.
- The permanent construction structure remains.
- Photos from other dates remain.
- Previously exported files remain untouched.

## 30. Final implementation principles

Prioritize:

1. Reliability.
2. Fast photo capture.
3. Clear hierarchy.
4. Offline operation.
5. Safe local file management.
6. Predictable report generation.
7. Simple Brazilian Portuguese terminology.
8. Low number of taps.
9. Readability in construction-site conditions.
10. Correct organization of exported files.

Do not expand the product into a general construction management platform.

The main success metric is:

An engineer can walk through the construction site, take hundreds of correctly classified photos and produce an organized PDF and ZIP without manually renaming, transferring or sorting any image.